



ON THE PHONEplain answer · how it is done · where the proof is

FEATURE	IN PLAIN WORDS	HOW IT IS DONE	PROOF
Scan a sculpture	Photograph a carving; the phone names it and tells its story with no internet.	EfficientNet-B0 (ImageNet-pretrained, 4.0 M parameters) fine-tuned to 25 classes; 14 epochs, AdamW + OneCycle, random-resized crop and colour jitter for cave light against harsh sun. Exported to ONNX (opset 17) and run in the browser by onnxruntime-web (WASM).	1,396 / 220 / 344 photos, split so no photographer is in both halves → 85.8% top-1, 98.0% site ml/vision/train.py
“Not sure” instead of a guess	If the photo is not one of the 25, it says so rather than inventing a name.	Open-set rule on the softmax: answer only when the top probability is at least 0.75, otherwise show candidates or answer at site level.	answers 79.4% at 95.6% accuracy ; rejects 81.8% of 33 Hampi / Belur / Halebidu photos ml/vision/ood_calibrate.py
Live lens (AR)	Hold the camera up and the name appears over the live view.	The same ONNX model on a timer loop over camera frames, exponential smoothing of the scores, a lock after three agreeing frames and hysteresis so the label does not flicker.	≈70 ms a frame in Chrome on a laptop (printed on screen) web/src/components/LiveLens.tsx
Read a board	Photograph an ASI board in any of three scripts and get the monument's story.	Tesseract.js (LSTM) with Kannada, Hindi and English traineddata in a web worker; the heading lines are matched against the knowledge base with heading weight ×3, stop-words removed and a site-level rule for overview boards.	4 / 4 real ASI boards matched to the right monument web/src/lib/ocr.ts
Translation	A whole board, or anything you type, in six languages.	Meta NLLB-200-distilled-600M on the district server: int8 dynamic quantisation, greedy decoding (one beam), a glossary that protects monument names, OCR-noise cleaning, streamed back sentence by sentence as NDJSON.	3.9× faster after int8; first line ≈ 6 s on a laptop CPU server/main.py · /api/translate
Ask the guide	Ask in Kannada, Hindi or English — typed, spoken or romanised.	17-intent classifier: character n-gram TF-IDF (2–4, char_wb) + logistic regression on the phone; multilingual-e5-small embeddings + logistic regression on the server; a curated trilingual lexicon on top. The answer is composed from tools (forecast, planner, parking, recommender) and a verified knowledge base. No LLM and no paid API .	95.8% server, 91.2% phone on unseen phrasings (400 train, 70 fresh test) ml/assistant/compare_intents.py
Voice in and out	Speak the question; the answer is read aloud.	Web Speech API for listening. For speaking, the phone's own voice where it has one, and Meta MMS-TTS (VITS: mms-tts-kan, mms-tts-hin) on the server for phones with no Kannada or Hindi voice.	web/src/lib/speech.ts server/main.py · /api/tts
Trip planner	Say your hours, budget and interests; get a timed day.	The sentence is parsed by rules and the lexicon; stops are chosen by interest and importance; then every visiting order is tried (exhaustive permutations) and scored on driving time, the forecast crowd at the time you would arrive, midday heat on stairs, opening hours and budget. A lunch stop is inserted where it fits.	the compute time is printed under the plan web/src/lib/itinerary.ts · tripParse.ts
Crowd forecast	How busy each site will be, hour by hour and day by day.	LightGBM (700 trees, learning rate 0.03, 31 leaves) on calendar features only: weekday, month, seasonal sine and cosine, Dasara, Diwali, Ugadi, Shivaratri, Sankranti, school holidays, long weekends and World Heritage Week. The 80% band comes from conformalised quantile regression.	WAPE 13.1% vs 31.1% same-day-last-year, on 243 unseen days of 2026; band holds 79.2% ml/forecast/train_footfall.py
Parking	How many spaces will be free when you actually arrive.	LightGBM (500 trees) per horizon — 30, 60 and 120 minutes — on occupancy history plus calendar features. A small 120-tree, 15-leaf copy is exported as JSON so the phone runs the same kind of model with no network.	±3.5 slots at 2 h vs ±7.9 for “no change”; 81% of “lot will be full” alerts right ml/forecast/train_parking.py
SOS, and SOS with no signal	Hold two seconds; it reaches the control room, or waits until the signal returns.	WebSocket to the district server. With no connection the alert goes into an outbox on the phone and is resent automatically on reconnect; the card falls back to a tel: call to 112 and an sms: with a maps link, neither of which needs data. Caution zones are haversine circles.	tested end to end with the network switched off web/src/lib/live.ts · pages/Safety.tsx
Stone check V1	Photograph a crack; the phone outlines crack-like marks for inspection.	Classical image processing, no model: each pixel is compared with its local average (dark-ridge filter), threshold at mean + 2.2 σ, then connected components are kept only when they are at least 40 px and three times longer than wide.	33 ms for a 720 px photo, on the phone web/src/lib/stone.ts
Stay, food and craft picks	Ranked for you, with the reason shown on every card.	Explainable weighted score: budget fit, distance to the places in your plan, amenities or cuisine, a Bayesian-smoothed rating, a small boost for locally owned places, then an MMR re-rank for variety.	19 stays, 34 eateries, names and places from OpenStreetMap web/src/lib/recommend.ts



IN THE COMMAND CENTRE

FEATURE	IN PLAIN WORDS	HOW IT IS DONE	PROOF
Live operations	Crowds, parking, heat and alerts on one screen.	The same forecast and parking models, plus live alerts pushed over a WebSocket; “respond” writes straight back to the visitor's phone.	server/main.py web/src/pages/Command.tsx
Carrying capacity V1	How many people a site can take in a day.	Comfort capacity × (open hours + average stay). Space per visitor and average stay are inputs on screen, so the department can use its own figures.	assumptions shown and editable pages/Command.tsx
Resource planning	Water, waste, toilets, guides, bins and trips for seven days.	Forecast × published-style rates: 4 L water and 0.12 kg waste per visitor, one toilet seat per 60 people at peak, one Pravasi Mitra per 150, a 120 L bin per 20 kg, twelve bins a trip. Every rate is editable.	pages/Command.tsx (constants at the top)
CCTV people counting	Counts people without anyone watching faces.	torchvision Faster R-CNN MobileNetV3-Large FPN (COCO weights), person class only, score ≥ 0.5, on the server. The frame is blurred in the browser and only the count and anonymous boxes are kept.	server/main.py components/CctvPanel.tsx
Sensor anomalies	Spots a stuck or lying sensor.	Robust z-score: median and median-absolute-deviation over the recent window for each device, so one bad reading does not move the baseline.	server/main.py · IoT ingest iot/ (ESP32 firmware)
What visitors say	Sentiment and complaint topics for every monument.	Character n-grams and a polarity lexicon on the phone; multilingual-e5-small + logistic regression on the server; complaint topics from a curated aspect lexicon (water, toilets, parking, shade, guides).	93.3% sentiment on 30 fresh reviews, topic F1 0.98 ml/assistant/

WHERE THE DATA COMES FROMall of it public, all of it credited

- Monument photographs — 1,960 images by 202 photographers, Wikimedia Commons (CC BY / BY-SA / CCO). Per-photo credit in ml/vision/data/metadata.csv
 - Places, roads, hospitals — OpenStreetMap contributors (ODbL) via Overpass API; drive times from OSRM (router.project-osrm.org)
 - Heritage facts — DTDC Bagalkote, *Sounds and Emotions* (2022), paraphrased and cited, plus bagalkot.nic.in/en/tourism
- Visitor numbers — ASI visitor footfall at centrally protected ticketed monuments, FY 2024–25: Badami 4,44,542 · Pattadakal 3,24,615 · Durga temple, Aihole 2,13,901
 - Weather — Open-Meteo forecast API, no key: temperature, apparent temperature and rain by latitude and longitude (open-meteo.com)
 - Open models — facebook/nllb-200-distilled-600M, facebook/mms-tts-kan · hin, intfloat/multilingual-e5-small, torchvision EfficientNet-B0 and Faster R-CNN, Tesseract kan · hin · eng

THE QUESTIONS YOU ARE ABOUT TO ASK

Where is the AI in this project?

Five models were **trained for Bagalkote**: the sculpture recogniser (EfficientNet-B0, on the phone), the footfall forecaster and the parking forecaster (LightGBM), the question classifier (char n-grams + lexicon + multilingual-e5) and the review sentiment and complaint-topic model. Five open models are used as they are: **NLLB-200** for translation, **MMS-TTS** for Kannada and Hindi speech, **Tesseract** for reading boards, **Faster R-CNN** for counting people, **multilingual-e5** for meaning. Around them sit classical algorithms: exhaustive route search, conformal prediction for the forecast band, robust z-scores for sensor faults, MMR for variety, and a dark-ridge filter for the crack check. **No LLM and no paid API**. Every model is measured against a simple baseline, so you can see what the AI adds. Training scripts and reports: [ml/*](#) and [ml/*out/](#).

How do you get the weather?

Open-Meteo's public forecast API by latitude and longitude, no key. We use the **apparent** temperature — what it feels like on hot stone — and warn above 38 °C.

How is traffic estimated?

It is not. Drive times come from OSRM routing on the OSM road network at typical speeds, cached per pair of sites. A live traffic feed is a version-2 item; I would rather say so than pretend.

How is the route chosen?

For the stops that fit your hours, **every order is tried** and scored on driving time, the crowd forecast at your arrival hour, midday heat on stairs, opening hours and budget.

Is the data real?

Photos, ticket totals, roads, places and weather: real. The hour-by-hour crowd shape, parking streams and business prices: **simulated and labelled in the app**, calibrated so yearly totals match ASI exactly.

What still works with no signal?

Recognition, live lens, board reading, planning, forecasts, parking, the knowledge base and SOS queueing. Translation, Grad-CAM and people counting need the district server.

What about privacy?

CCTV frames are blurred and only counts are stored. Questions are counted anonymously. The emergency contact never leaves the phone. No accounts, no tracking, no ads.

WHAT RUNS WHEREthe honest answer to “does it work offline?”

On the phone

works with no signal, after the first visit

Sculpture model (ONNX) · live lens · board OCR (Tesseract) · trip planner · crowd forecast · parking model (120 trees as JSON) · knowledge base · stone check · SOS outbox · everything cached by the service worker

On the district server

one laptop today, one small machine for the district

Live alerts over WebSocket · NLLB-200 translation (int8) · MMS-TTS speech · Grad-CAM “why” · CCTV people counting · sensor anomaly checks · multilingual-e5 for questions and reviews

Public services

free, no keys, no accounts

Open-Meteo weather (live) · OpenStreetMap tiles for the maps · OSRM drive times (fetched once, cached in the app) · Wikimedia Commons photographs (fetched when the dataset was built)